OO-S-256/15A
June 13, 1986
SUPERSEDING
OO-S-00256(GL)/15
June 11, 1985

FEDERAL SPECIFICATION SHEET

SEWING MACHINE, INDUSTRIAL; FLATBED, DOUBLE NEEDLE, 301 STITCH, PARACHUTE REPAIR

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration for the use of all Federal Agencies.

The complete requirements for procuring the sewing machines described herein shall consist of this document and the issue in effect of Specification OO-S-256.

1. SCOPE

1.1 Scope. This specification covers the specific requirements for double needle, flatbed, 301 stitch, heavy duty, industrial sewing machines used in the fabrication and repair of aerial delivery equipment.

2. APPLICABLE DOCUMENTS

2.1 Government publications. The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

6-1-9366 Sewing Machine, Industrial, Flatbed, Double Needle, 301 Stitch Heavy Duty: Parachute Repair and Fabrication Test Requirements

AMSC N/A FSC 3530

DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

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(Copies of Military specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Machine, heavy duty. The heavy duty machine shall be reversible with the capability of sewing light to heavy cotton and nylon duck, light to heavy tape and webbing and a combination of any of the above utilizing metric needle sizes 110, 120, 125, 130, and 140. The machine shall operate at speeds not less than 1500 or greater than 2500 stitches per minute. The machine shall be manually or semi-automatically lubricated. The machine shall have a split needle bar with a disengageable right and left needle bars. Disengagement shall be achieved by an activating control. The mounting shall be fully submerged. The machine shall be equipped with a bobbin winder and a knee lifter.

3.1.1 Machine feed. The machine shall have a reversible type H feed. The feed dogs shall have three rows of teeth. They shall include one full row of teeth between the needle openings and one partial row in the front and one partial row in the rear of each of the needle openings.

3.1.2 Machine, stitches. The machine shall produce a minimum of 4 to a unlimited number of stitches per inch.

3.1.3 Needles, distance between. The distance between the needles shall be adjustable from 1/8 of an inch to 1 inch.

3.1.4 Machine, height under the presser foot. The machine shall provide minimum height of 11/32 of an inch.

3.1.5 Machine, working room to the right of the needle. The machine shall be furnished with a working room distance to the right of the needle of 10 inches minimum to 16 inches maximum.

3.1.6 Machine, table top. The machine shall be furnished with a type II table top. The outside dimensions of the table top shall be 20 by 40 by 2 inches thick with a tolerance of ± 1/4 of an inch for each dimension.

3.1.7 Machine gauge sets. The machine shall be supplied with complete gauge sets for sizes 1/8, 1/4, 3/8, 1/2 and 3/4 of an inch. Each set shall

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include vibrating presser foot, lifting presser foot, needle clamp, needle throat plate, feed dog and any other component or special tool required to perform each double needle stitch formation. The 3/8 inch gauge set shall be installed in the machine.

3.1.8 Spare parts. The machine shall be furnished with the following spare parts in the quantity indicated:

Spare parts

Item identification	Quantity
Face plate screws	2
Thread take up spring	1
Thread tension thumb nut	1
Hook assembly sewing	1
Hook holding screw sewing	1
Bobbin, metal	10
Bobbin case assembly	1
Bed slide plate spring	1
Bed slide plate spring screw	1
Needle throat plate screw	2
Needle set screw	2
Needle clamps for 3/8 of an inch	2
Needle clamps for 1/2 of an inch	2
Needle clamp set screw	2
Needle bar thread guides	2
Needle bar thread bar screw	2
Presser foot holding screw	2
Needle, round point, chrome finish	
Metric sizes:	
110	100
120	100
125	100
130	100
140	100

3.2 Performance.

3.2.1 Machine operation. The heavy duty machine shall be capable of fabricating, repairing, and installing sections with lap seams on personnel and cargo parachutes when tested as specified in 4.1.1 and 4.2.1. The machine shall show no signs of stalling, pounding, shearing of threads, failure to lock stitches, or deformation of any part during the fabrication of these articles.

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4. QUALITY ASSURANCE PROVISIONS

4.1 First article testing. When a first article is required it shall be tested in accordance with the applicable test method specified in table I.

TABLE I. First article tests

Sewing machine	Requirement paragraph	Test method
Heavy duty	3.2.1	4.1.1

4.1.1 Machine test. The machine shall be operated to fabricate and assemble all of the typical parachute details as shown on Drawing 6-1-9366. Any nonconformance with the requirements of 3.2.1 shall constitute failure of this test.

4.2 End item testing. Each end item shall be tested in accordance with the applicable test method specified in table II. Failure of any test shall be cause for rejection of the end item involved.

TABLE II. End item testing

Sewing machine	Requirement paragraph	Test method
Heavy duty	3.2.1	4.2.1

4.2.1 Machine test. Each machine shall be operated to sew samples of typical parachute construction, as shown by details F and G on Drawing 6-1-9366. Any nonconformance with the requirements of 3.2.1 shall constitute failure of this test. The completed test samples shall be supplied with each machine furnished.

5. PACKAGING

Packaging shall be as specified in the basic specification and in the contract or purchase order.

6. NOTES

6.1 Intended use. The sewing machines are intended for industrial use for fabricating and repairing parachutes.

6.2 Ordering data.

Title, number, and date of this specification.

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MILITARY INTERESTS:

Custodians:

Army - GL
Air Force - 99

Review activity:

DLA-GS

User activity:

Air Force - 50

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

Preparing activity:

Army - GL

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